

LC-3 Instruction Encodings

Arith./Logic	ADD [†]	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
		0	0	0	1	DR			SR1			0			0	0	SR2							
	ADD [†]	0	0	0	1	DR			SR1			1	imm5											
	AND [†]	0	1	0	1	DR			SR1			0			0	0	SR2							
	AND [†]	0	1	0	1	DR			SR1			1	imm5											
	NOT [†]	1	0	0	1	DR			SR			1	1	1	1	1	1							
Load/Store	LD [†]	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
		0	0	1	0	DR			PCoffset9															
	ST	0	0	1	1	SR			PCoffset9															
	LDI [†]	1	0	1	0	DR			PCoffset9															
	STI	1	0	1	1	SR			PCoffset9															
	LDR [†]	0	1	1	0	DR			BaseR			offset6												
	STR	0	1	1	1	SR			BaseR			offset6												
Branch	JMP	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
		1	1	0	0	0			0	0	BaseR			0						0	0	0	0	0
	BR	0	0	0	0	cn	cz	cp	PCoffset9															
	JSRR	0	1	0	0	0			0	0	BaseR			0						0	0	0	0	0
	JSR	0	1	0	0	1	PCoffset11																	
Misc.	LEA	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
		1	1	1	0	DR			PCoffset9															
	TRAP	1	1	1	1	0			0	0	0	0	trapvect8											
	RTI	1	0	0	0	0												0	0	0	0	0	0	0
	<i>illegal</i>	1	1	0	1	x	x	x	x	x	x	x	x	x	x	x	x							

[†] Updates condition code (CC)

LC-3 MMIO Registers

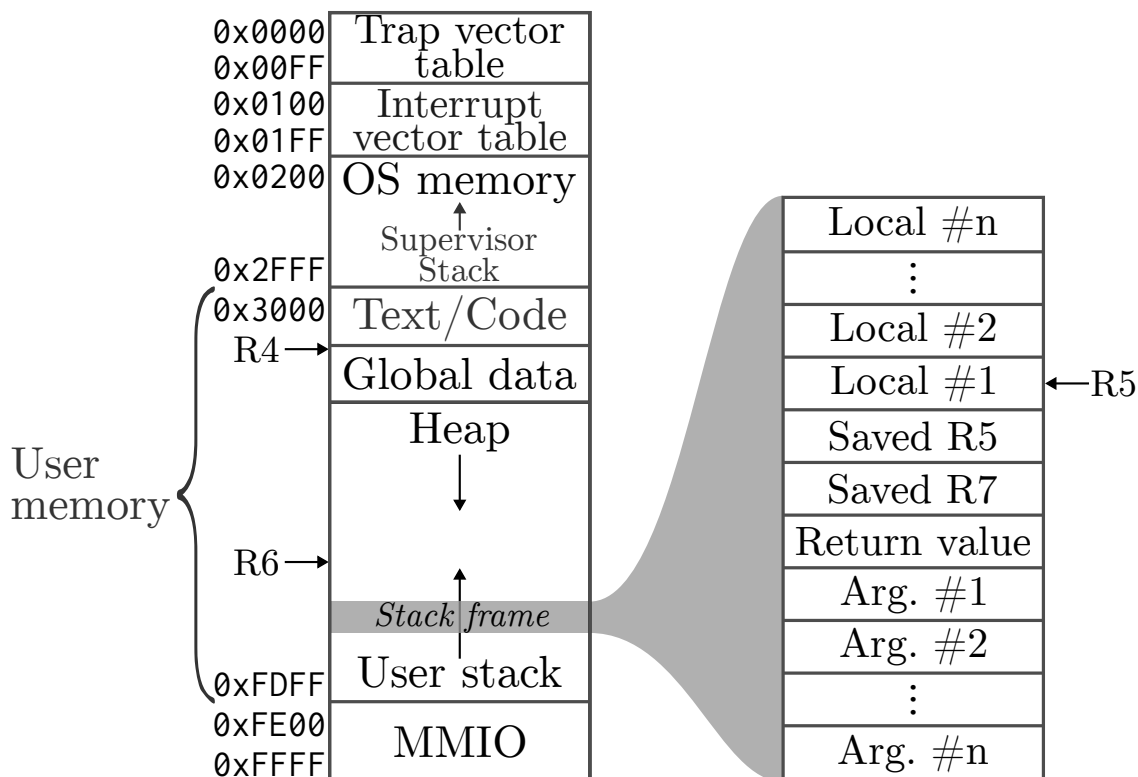
		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xFE00	KBSR	Rdy	(undefined)														
0xFE02	KBDR	0 0 0 0 0 0 0 0								ASCII code							
0xFE04	DSR	Rdy	(undefined)														
0xFE06	DDR	0 0 0 0 0 0 0 0								ASCII code							
0xFFFC	PSR	Priv	0 0 0 0				PRIO			0 0 0 0 0				N	Z	P	

(Priv = 0 is supervisor mode. PRIO = 111₂ is highest.)

LC-3 Trap Vectors

- 0x20 → GETC
- 0x21 → PUTC
- 0x22 → PUTS
- 0x25 → HALT

LC-3 Memory Map



LC-3 Datapath Control Signals

(Use these for filling out LC-3 FSM *state transition diagrams*)

Signals to assert:

- LD.MAR
- LD.MDR
- LD.IR
- LD.REG
- LD.CC
- GatePC
- GateMDR
- GateALU
- GateMARMUX
- GateSP
- GatePSR
- MIO.EN
- ACK

Signals with named values:

- **PCMUX**: PC+1, BUS, ADDER
- **SR1MUX**: IR[11:9], IR[8:6], 6
- **DRMUX**: IR[11:9], 7, 6
- **ALUK**: ADD, AND, NOTA, PASSA
- **ADDR1MUX**: PC, BaseR
- **ADDR2MUX**: ZERO, offset6, PCoffset9, PCoffset11
- **MARMUX**: IR[7:0], ADDER
- **SPMUX**: Saved.USB, Saved.SSP
- **MIO.RW**: R, W

LC-3 Microcode Control Signal Bit Assignments

(Use these for filling out bits in *microcode tables*)

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|--|--|
| <ul style="list-style-type: none"> • PCMUX: <ul style="list-style-type: none"> • 00₂ → PC+1 • 01₂ → BUS • 10₂ → ADDER • SR1MUX: <ul style="list-style-type: none"> • 00₂ → IR[11:9] • 01₂ → IR[8:6] • 10₂ → 6 • DRMUX: <ul style="list-style-type: none"> • 00₂ → IR[11:9] • 01₂ → 7 • 10₂ → 6 • ALUK: <ul style="list-style-type: none"> • 00₂ → ADD • 01₂ → AND • 10₂ → NOTA • 11₂ → PASSA | <ul style="list-style-type: none"> • SPMUX: <ul style="list-style-type: none"> • 0 → Saved.USB • 1 → Saved.SSP • ADDR1MUX: <ul style="list-style-type: none"> • 0 → PC • 1 → BaseR • ADDR2MUX: <ul style="list-style-type: none"> • 00₂ → ZERO • 01₂ → offset6 • 10₂ → PCoffset9 • 11₂ → PCoffset11 • MARMUX: <ul style="list-style-type: none"> • 0 → IR[7:0] • 1 → ADDER • MIO.RW: <ul style="list-style-type: none"> • 0 → R • 1 → W |
|--|--|

